The Observability Calculus

A calculus for tracking uncertainty from sensor to estimate

Contents

1. Introduction

The calculus applies to any chain of observation derived from transducing sensors. It tracks uncertainty from sensor to downstream estimate.

It is especially useful for DOME because the chains are long, cascading, and integrate heterogeneous sensors: a downstream estimate such as retinal input is only as trustworthy as the entire chain that produces it.

Its use is to decide whether a buildable instrument provides sufficient precision, accuracy, and reliability to answer a given research question, and, where it does not, to identify the step that limits it.

1.1. The chain

- **Energy.** Physical energy the environment makes available: light, inertial, contact force. A quantity carried by no available energy is unmeasurable.
- **Transduction.** A sensor converts one energy to a raw signal (cameras, IMUs, force plates). The one irreversible step. Each sensor fixes the reference frame of its raw signal.
- **Cascade.** Typed derivations: pose detection, triangulation, rigidification, composition, fusion, projection.
- **Estimate.** The chain terminates at the desiderata — the signals nearest the nervous system, in the frame the nervous system uses (retinal input, retinotopic; muscle drive).

1.2. Steps typed by their effect on uncertainty

Every step is classified by what it does to uncertainty; this is the core of the calculus. Two error kinds are tracked separately: *bias* (systematic error that repeats and does not average away) and *variance* (random error that averages away over repeats).

Effect	Step	Instances
Preserve	exact computation	triangulation, coordinate transform, rigid-body kinematics given parameters
Amplify	map with gain	projection: existing error scaled by the Jacobian (gaze-point error = gaze-angle error × depth)
Inject variance	stochastic inference	ML pose detection, learned priors, redundancy resolution; partly reported (confidence), partly not
Inject bias	assumption / injected constant	rigid-body assumption (error $\propto$ deviation from rigidity); anthropometry tables (error $\propto$ table quality × subject mismatch); usually unreported

Preserve and amplify add no new uncertainty; they only transform error already present. Inject-variance and inject-bias add new error, by different mechanisms. Composition and fusion combine estimates and add error only through miscalibration.

2. Type system

2.1. Frames

Definition 1 (reference frame). A *reference frame* F is a coordinate system fixed either by a sensor (its physical geometry) or by a *registration procedure*: a construction that expresses one sensor's frame in a shared world (inertial) frame. *In words: the coordinate system a number is measured in; two numbers are only comparable once put in the same frame.*

2.2. Channels and values

Definition 2 (channel). A *channel* c is one degree of freedom, with value space V_c, belonging to a frame $F(c)$. *In words: one independent quantity — e.g. one rotation axis of the eye.*

Definition 3 (value). A *value* is a tuple $v = (v_c)_{c \in C}$ over a channel set C, with $\dim v = |C|$. *In words: a bundle of channels; status is tracked per channel, never as one verdict on the whole bundle.*

2.3. Estimates

Definition 4 (estimate). An *estimate* of a value v with true latent v^* is a triple $\hat{e} = (\hat{v}, b, \Sigma)$ in a frame F, where

$$e = \hat{v} - v^* = b + n, \quad \mathbb{E}[n] = 0, \quad \mathrm{Cov}(n) = \Sigma.$$

In words: $\hat{v}$ is our estimate, v^* the truth we want, e the error; b is the bias (systematic, repeats), n the noise (random, averages out), and Σ the noise's covariance (its size and its correlations across channels). Bias and covariance are tracked separately because different step types produce different ones.

2.4. Per-channel status

The status of channel c is the marginal (b_c, Σ_{cc}). *In words: how far off channel c is on average (b_c) and how noisy it is (Σ_{cc}, the diagonal entry).* It is a magnitude, not a flag: *well measured* means both small; *inferred / prior-dominated* means Σ_{cc} large or b_c set by an assumption. The off-diagonal Σ_{cd} is the noise shared between channels c and d.

2.5. Quantified vs. unquantified uncertainty

Both terms split into a reported and an unreported part, $b = b_q + b_u$ and $\Sigma = \Sigma_q + \Sigma_u$. *In words: subscript q is the part we can quote a number for; subscript u is the part we know is there but cannot size.* Computation contributes to neither; inference contributes Σ_q (reported confidence / posterior) and Σ_u (unreported); an assumption contributes b_u (bias present, magnitude unknown), and b_q only where the deviation is measured. The u parts are carried so an unreported error is never treated as zero.

2.6. Inter-frame transforms and calibration

Definition 5 (transform, calibration). A *transform* $T : F \to G$ is itself an estimate — a pose value with its own error (b_T, Σ_T) — carrying a *calibration* quality

$$\kappa = (\kappa_s, \kappa_t) \in [0, 1]^2.$$

In words: how well two instruments are aligned, from 0 (independent) to 1 (perfect), split into κ_s spatial (geometric registration) and κ_t temporal (synchronization). $\kappa = (1, 1) \Rightarrow T$ exact ($b_T = 0, \Sigma_T = 0$); $\kappa < 1 \Rightarrow (b_T, \Sigma_T) > 0$, growing as $\kappa \to 0$.

Mapping a value through T composes T's error into it; this is the only path by which miscalibration enters. κ_s and κ_t degrade independently.

3. Operators

An *operator* is a step in the pipeline: a map on estimates. We linearize it at the operating point $\hat{v}$ and write $J = \partial O / \partial v$ for its *Jacobian* (*In words: how much and in what direction the output moves when the input moves; its size $\| J \|$ is the gain*).

Definition 6 (general operator). O acts as

$$(\hat{v}, b, \Sigma) \longmapsto \left(O\hat{v},\ Jb + d,\ J\Sigma J^\top + Q\right),$$

with d the *injected bias* and $Q \succeq 0$ the *injected covariance* (*In words: the fresh bias and fresh noise this step adds on top of transporting what came in; $\succeq 0$ means a valid, non-negative amount of noise*). The pair (d, Q) classifies O.

Bias convention. $b = b_q + b_u$: b_q propagates as a vector ($b_q \mapsto Jb_q$); b_u is carried as a bound $\| b_u \| \leq \beta$ with $\beta \mapsto \| J \| \beta$. *In words: known bias is pushed through like a value; unknown bias is tracked as a worst-case size.*

3.1. The three mechanisms

Definition 7 (transport). $d = 0,\ Q = 0$: $(\hat{v}, b, \Sigma) \mapsto \left(O\hat{v}, Jb, J\Sigma J^\top\right)$. Adds nothing; existing error is carried through J. *Amplifies* where $\| J \| > 1$, *contracts* where $\| J \| < 1$. Exact computation and geometric projection are both transport; projection is the large-$\| J \|$ case (error scaled by depth).

Definition 8 (inject variance). $d = 0,\ Q = Q_q + Q_u \succeq 0$: $\Sigma \mapsto J\Sigma J^\top + Q$. Stochastic inference. *In words: a model-based step that adds noise; part reported, part not.*

Definition 9 (inject bias). $Q = 0,\ d = d_q + d_u$ with $\| d_u \| \leq \delta$: $b \mapsto Jb + d$. An assumption or injected constant. *In words: adds a systematic offset whose worst-case size δ is set by how wrong the assumption is for this subject.*

3.2. Composition

Definition 10 (composition $\oplus$). For an estimate $(\hat{v}, b_v, \Sigma_v)$ carried across frames by a transform T, with Jacobians J_v, J_T,

$$b_w = J_v b_v + J_T b_T, \quad \Sigma_w = J_v \Sigma_v J_v^\top + J_T \Sigma_T J_T^\top.$$

In words: chaining across frames transports the value and adds the transform's own calibration error. At $\kappa = (1, 1)$ it is pure transport.

3.3. Fusion

Definition 11 (fusion $\bowtie$, independent case). For two estimates of the same value with *independent* errors,

$$\Sigma_f = \left(\Sigma_1^{-1} + \Sigma_2^{-1}\right)^{-1}, \quad \hat{v}_f = \Sigma_f(\Sigma_1^{-1}\hat{v}_1 + \Sigma_2^{-1}\hat{v}_2), \quad b_f = \Sigma_f(\Sigma_1^{-1}b_1 + \Sigma_2^{-1}b_2).$$

In words: combine two readings of one thing, weighting each by its certainty (Σ^{-1}).

Proposition 1. $\Sigma_f \preceq \Sigma_1$ and $\Sigma_f \preceq \Sigma_2$. (*In words: fusing never increases uncertainty; $\preceq$ is the Loewner order, "at most as much covariance in every direction."*)

Fusion injects nothing itself under independence and a common frame. When the two errors are correlated (a shared calibration), the independent formula understates the error; the general correlated form is in the Chain-propagation section.

3.4. Summary

Operator	Mechanism	Injected (d, Q)
computation, projection	transport	$(0, 0)$; error carried by J
inference	inject variance	$(0, Q)$, $Q \succeq 0$
assumption / constant	inject bias	$(d, 0)$, $\| d_u \| \leq \delta$
composition $\oplus$	transport + calibration	$\left(J_T b_T,\ J_T \Sigma_T J_T^\top\right) \to 0$ as $\kappa \to 1$
fusion $\bowtie$	variance reduction	none (independent, common frame)

4. Chain propagation

4.1. The pipeline as a graph

A pipeline is a directed acyclic graph $\mathcal{D} = (N, E)$: nodes N are estimates, edges E are operators. *In words: a wiring diagram of the whole measurement chain.* An edge $e : i \to j$ carries (J_e, d_e, Q_e) and acts as in the general operator. Each injection is a fresh *primitive source* ξ_e: an independent original error with mean μ_e (its bias) and covariance Q_e (its noise). Stack them into ξ, with mean μ and block-diagonal covariance $W = \mathrm{Cov}(\xi)$. *In words: every bit of new error entering anywhere gets its own independent "source"; because they are independent, W has no cross terms.*

4.2. Linear error model

Definition 12 (sensitivity). $S_i = \sum_{\text{paths } p:\cdot\rightsquigarrow i} \prod_{e\in p} J_e$, the total sensitivity of node i's error to the primitives. *In words: add up every route each source can take to reach node i, multiplying the gains along each route.*

Proposition 2 (closed form). $e_i = S_i\xi$, so $b_i = S_i\mu$ and $\Sigma_i = S_i W S_i^\top$. *In words: a node's bias is the sum of source biases routed to it; its covariance is the sum of source noises routed to it.*

The edge-by-edge recursion is the operator law together with $\Sigma_{jk} = J_e\Sigma_{ik}$ for any other node k (the fresh primitive at e is independent of k).

4.3. Cross-covariance and shared sources

Definition 13 (cross-covariance). $\Sigma_{ij} = S_i W S_j^\top$. *In words: how much two nodes' errors move together.* Since W is block diagonal, $\Sigma_{ij} \neq 0$ iff i and j inherit a common primitive.

A single calibration feeding two routes is one such primitive, so the two routes are correlated; tracking Σ_{ij} where routes converge is mandatory.

4.4. Fusion under correlation

With $P_{11} = \Sigma_i,\ P_{22} = \Sigma_j,\ P_{12} = \Sigma_{ij} = P_{21}^\top$ and $M = P_{11} + P_{22} - P_{12} - P_{21}$,

$$\hat{v}_f = \hat{v}_i + (P_{11} - P_{12})M^{-1}(\hat{v}_j - \hat{v}_i), \quad \Sigma_f = P_{11} - (P_{11} - P_{12})M^{-1}(P_{11} - P_{21}).$$

In words: the general two-estimate fusion that accounts for shared error; reduces to the independent formula when $P_{12} = 0$.

Proposition 3 (shared error is not attenuated). Write $e_i = c + n_i,\ e_j = c + n_j$ with c the common part (shared primitives) and n_i, n_j independent. Then fusion leaves c intact and $\Sigma_f \succeq \mathrm{Cov}(c)$. *In words: the error two routes share — e.g. a common calibration bias — cannot be averaged away; it sets a floor.*

4.5. Terminal result

At a terminal node t (a desideratum) and channel c, propagation returns the marginal $(b_{t,c}, \Sigma_{t,cc})$, each with its reported/unreported split; the unreported-bias bound propagates as $\beta_j = \| J_e \| \beta_i + \delta_e$, maximized over converging paths. *In words: the final accuracy and precision delivered on the thing we actually wanted.*

5. The sufficiency test

Definition 14 (requirement). A research question fixes a set R of required channels and, for each $c \in R$, tolerances b_c^* (accuracy) and σ_c^{*2} (precision). *In words: which quantities the study needs, and how accurate and how precise each must be.*

Definition 15 (worst-case delivered).

$$\overline{b}_c = \| b_{q,c} \| + \beta_c, \quad \overline{\Sigma}_c = \Sigma_{q,cc} + \Sigma_{u,cc},$$

with $+\infty$ where a required bound is missing. *In words: the delivered error counting the unknown parts at their worst; ∞ if a part cannot be bounded at all.*

Definition 16 (margins and verdict). $m_c^b = b_c^* - \overline{b}_c$ and $m_c^\Sigma = \sigma_c^{*2} - \overline{\Sigma}_c$. For each $c \in R$: **sufficient** if both margins ≥ 0 with finite bounds; **insufficient** if a margin < 0 with finite bounds; **uncertifiable** if a required bound is infinite. The system is sufficient for the question iff every $c \in R$ is sufficient. The *binding channel* is $\arg\min_{c\in R} \min(m_c^b, m_c^\Sigma)$. *In words: pass if delivered beats required on both accuracy and precision; fail if it loses; "uncertifiable" if an unknown-magnitude error means we cannot say either way. The binding channel is the one with the least slack.*

5.1. Deficit attribution

Both delivered quantities are additive over the primitive sources. With u_c the selector for channel c and $a_{c,s}$ the block of $S_t^\top u_c$ belonging to source s,

$$\Sigma_{t,cc} = \sum_s \rho^\Sigma_{c,s},\ \ \rho^\Sigma_{c,s} = a_{c,s}^\top W_s a_{c,s}; \quad b_{t,c} = \sum_s \rho^b_{c,s},\ \ \rho^b_{c,s} = a_{c,s}^\top \mu_s.$$

In words: split the delivered noise and bias on channel c into the share coming from each source.

Definition 17 (development target). On the binding channel c, $s^* = \arg\max_s \rho_{c,s}$ on the violated term. *In words: the single source — a tracker, a calibration, an assumption, an injected constant — contributing most to the shortfall, i.e. the thing to improve first.*

5.2. Examples

Fine bark texture during navigation: required precision is below any $\overline{\Sigma}_c$ the sensors deliver; $m_c^\Sigma < 0$; insufficient, with s^* the imaging sensor's resolution. Steering around a tree: on the required body-position, heading, and gaze channels both margins are ≥ 0; sufficient.

6. Feedback and validation

These constructions run against the forward chain: they use a downstream estimate, or an outside reference, to reduce an injected error or to bound an unreported one. Each changes either W (the sizes of the original error sources) or the bounds on unreported error, and the delivered numbers follow.

6.1. Back-projection

Let m be an inference node (e.g. a learned pose detector) injecting Q_m, and t a downstream estimate that pools more information (fusion across sensors and time). Let h be the forward model from state to m's measurement space, with Jacobian $P = \partial h / \partial v$. *In words: P says how the thing m measures depends on the state.*

Definition 18 (reprojection residual). $r = \hat{v}_m - h(\hat{v}_t)$. *In words: what the detector reported minus what the pooled downstream estimate predicts it should have reported.*

Retraining m on the target $h(\hat{v}_t)$ reduces its injected covariance where the pooled prediction is more certain than the raw detector.

Definition 19 (reducible subspace). S_{bp} is spanned by the directions where $P \Sigma_t P^\top \prec Q_m$. *In words: the directions in which the downstream estimate beats the detector.*

Proposition 4 (observability guard). Back-projection can lower Q_m only on S_{bp}; off it, $P\Sigma_t P^\top \succeq Q_m$ and using $h(\hat{v}_t)$ as target would inject the pooled estimate's error instead. The update gives $Q_{m'} \preceq Q_m$, with equality off S_{bp}.

Since t is downstream of m, the two are correlated (Σ_t contains m's error). The usable improvement comes only from primitives other than m; if m is t's sole source, $P\Sigma_t P^\top$ contains Q_m in full and there is no gain. *In words: a detector cannot bootstrap off itself — the gain is exactly the information the other sensors and time add.*

Proposition 5 (fixpoint). Replacing Q_m by $Q_{m'}$ gives $W' \preceq W$, hence $\Sigma_{t'} = S_t W' S_t^\top \preceq \Sigma_t$; iterating improves the target and lowers Q_m again, converging to a fixpoint bounded by the non-m information.

6.2. Cleaning versus predicting

Both learn a map from available channels a to a target channel τ. They differ by whether τ was measured.

Definition 20 (cleaning). τ is measured (finite Q_τ). A model of clean signals (e.g. flagship-trained) is the target, reducing $Q_\tau \mapsto Q_{\tau'} \preceq Q_\tau$ on the subspace where that model is more certain. This is back-projection with an external teacher; the output stays a measurement. *In words: denoise a channel you did measure; still a measurement.*

Definition 21 (predicting). τ is unmeasured (its column in S_t is zero). A learned edge injects $\hat{\tau} = \mathbb{E}[\tau \mid a]$ as a new inference, with residual covariance

$$\mathrm{Var}(\tau \mid a) = \mathrm{Var}(\tau) e^{-2I(\tau;a)},$$

where $I(\tau; a)$ is the mutual information between target and available channels. *In words: fill a channel you never measured from correlated ones; you recover it only up to how much they share, and the result is model-generated, not a measurement.*

Proposition 6. Predicting enters the propagation as inject-variance carrying its own unreported part; it cannot certify τ unless the learned conditional is itself validated. Its ceiling is $I(\tau; a)$: no map from a recovers τ below $\mathrm{Var}(\tau)e^{-2I(\tau;a)}$.

Cleaning serves measurement; predicting serves uses where a plausible value, not a measured one, suffices (e.g. behaviour modelling, inverse reinforcement learning).

6.3. Validation

Let a reference supply $\hat{v}_{\mathrm{ref}}$ with error $(b_{\mathrm{ref}}, \Sigma_{\mathrm{ref}})$ and observable subspace range S_{ref}. Let $\Pi_\cap$ project onto the shared channels range $S_t \cap$ range S_{ref}. *In words: only channels both the system and the reference observe can be compared.*

Definition 22 (validation residual and test). $r = \Pi_\cap(\hat{v}_t - \hat{v}_{\mathrm{ref}})$, with $\mathrm{Cov}(r) = \Pi_\cap\left(\Sigma_t + \Sigma_{\mathrm{ref}} - \Sigma_{t,\mathrm{ref}} - \Sigma_{t,\mathrm{ref}}^\top\right)\Pi_\cap^\top$. Under the null (both unbiased, correctly modelled),

$$r^\top \mathrm{Cov}(r)^{-1} r \sim \chi^2(\dim \Pi_\cap),$$

and a bias b appears as noncentrality $b^\top \mathrm{Cov}(r)^{-1} b$.

- **External.** Reference is a gold standard (Vicon): $b_{\mathrm{ref}} \approx 0$, Σ_{ref} small and known. Certifies the system's bias on $\Pi_\cap$.
- **Cross.** Reference is a second instrument (small DOME in large DOME) with its own $(b_{\mathrm{ref}}, \Sigma_{\mathrm{ref}})$; correlated iff the two share primitives (a common calibration object), so $\Sigma_{t,\mathrm{ref}}$ is required. Each refers the other on $\Pi_\cap$.
- **Use-based.** Reference is a theoretical prediction from a standard experiment (gait; base-of-support vs. centre-of-mass; CTSIB-M). The prediction carries the theory's own error $(b_{\mathrm{th}}, \Sigma_{\mathrm{th}})$ (e.g. anthropometry), so the residual tests instrument and theory jointly; a pass certifies only up to the theory's error.

Proposition 7 (validation resolves "uncertifiable"). A passed validation yields a finite empirical bound $\| b_{t,c} \| \leq \| r \| +$(reference bound) on $\Pi_\cap$, replacing an infinite bound on an unreported part. The channel's margin becomes computable, moving its verdict from uncertifiable to sufficient or insufficient.

6.4. Coupling to sufficiency

Let $\mathcal{F}$ be the set of applied feedback constructions. The propagation is evaluated on the post-feedback $W(\mathcal{F})$ and bounds $(\mathcal{F})$, so the verdict is a function of $\mathcal{F}$. *In words: what counts as "sufficient" depends on what feedback you have already applied; sufficiency is a property of the system-plus-feedback, not the raw chain.*

- **Insufficient.** Back-projection or cleaning lowers the injected covariance at the development target s^*, raising the binding channel's precision margin; this can flip the verdict without new hardware. Where it cannot (the reducible subspace excludes the deficient direction), the deficit is genuinely a hardware limit.
- **Uncertifiable.** Validation bounds the offending unreported part, making the margin computable.

- **Predicting** adds channels but with their own unreported error; it does not certify, and applies to questions whose requirement is plausibility rather than measurement.

The order follows: reduce where reducible (back-projection, cleaning), bound where unbounded (validation), and read the residual deficit as the true hardware target.

7. Symbol reference

Symbol	Plain-English meaning
v	a value: a bundle of numbers, one per degree of freedom
v^*	the true value we want (the hidden ground truth)
$\hat{v}$	our estimate of the value
e	error: estimate minus truth, $\hat{v} - v^*$
b	bias: the systematic part of the error (repeats every time; does not average away)
n	noise: the random part of the error (zero mean; averages away over repeats)
Σ	covariance: how large the noise is and how it correlates across channels
Σ_{cc}	variance of channel c (a diagonal entry): how noisy that one channel is
Σ_{cd}	covariance between channels c and d: their shared noise
b_q, Σ_q	the reported (quantified) parts of bias and noise
b_u, Σ_u	the unreported parts, known to exist but of unknown size
β	a worst-case bound on the size of the unreported bias
δ	a worst-case bound on the bias an assumption injects
c	a channel: one degree of freedom
V_c	the space channel c's value lives in
C	the set of channels that make up a value
F, G	reference frames (coordinate systems)
$F(c)$	the frame channel c is expressed in
T	a transform between frames (itself an uncertain pose)
$\kappa = (\kappa_s, \kappa_t)$	calibration quality in $[0, 1]$: 0 uncalibrated, 1 perfect; spatial and temporal
O	an operator: one step in the pipeline
J	Jacobian: how much / which way the output moves per unit input move (the local gain)
$\parallel J \parallel$	the size of that gain (> 1 amplifies error, < 1 shrinks it)
d	injected bias: fresh systematic error a step adds
Q	injected covariance: fresh noise a step adds
$\oplus$	compose: chain estimates across frames
$\bowtie$	fuse: combine two estimates of the same quantity
$\mathcal{D} = (N, E)$	the pipeline as a graph: nodes are estimates, edges are operators
ξ	the primitive sources: the independent original injected errors
W	covariance of the primitives (block-diagonal, since they are independent)
μ	means of the primitives (their biases)
S_i	sensitivity of node i: how each source reaches i (paths of Jacobian products)
Σ_{ij}	cross-covariance of nodes i, j (nonzero iff they share a source)
M	the correction denominator for fusing two correlated estimates
t	a terminal node: a desideratum, the quantity we ultimately want
u_c	selector picking out channel c
R	the channels a research question requires

Symbol	Plain-English meaning
b^*_c, σ^{*2}_c	required accuracy and precision on channel c
$\overline{b}_c, \overline{\Sigma}_c$	worst-case delivered accuracy and precision (unknown parts bounded in)
m^b_c, m^Σ_c	margins: required minus delivered (≥ 0 means pass)
$\rho^\Sigma_{c,s}, \rho^b_{c,s}$	how much source s contributes to channel c's variance / bias
s^*	the development target: the source contributing most to the deficit
P	forward-model Jacobian: how a measured quantity depends on the state (back-projection)
r	residual: measured minus predicted, or system minus reference
$Q_m, Q_{m'}$	an inference node's injected covariance, before and after back-projection
S_{bp}	reducible subspace: directions where the pooled estimate beats the detector
$I(\tau; a)$	mutual information: how much the available channels a reveal about target τ
τ, a	a target channel to fill, and the available channels used to fill it
$\Pi_\cap$	projection onto the channels both the system and the reference observe
$\hat{v}_{\mathrm{ref}}$	the reference estimate, with error $(b_{\mathrm{ref}}, \Sigma_{\mathrm{ref}})$
χ^2	chi-squared distribution, used as the validation agreement test
$\mathcal{F}$	the set of applied feedback constructions
$\preceq$	Loewner order: $A \preceq B$ means B has at least as much covariance as A in every direction
$\succeq 0$	positive semidefinite: a valid covariance (never negative variance)
$\mathbb{E}[\cdot]$	expected value (the average over repeats)
$\mathrm{Cov}(\cdot)$	covariance